

CNN Training Report

Image Classification with Convolutional Neural Networks

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February 17, 2026

1 Introduction

This report documents the design, implementation, and evaluation of a Convolutional Neural Network (CNN) for image classification tasks. Two models were developed: Model 1 for 10-class digit recognition (Dataset 1) and Model 2 for 100-class object recognition (Dataset 2).

2 Model Architecture Design and Rationale

2.1 Model 1: 10-Class Digit Recognition

2.1.1 Architecture

- **Input Layer:** Grayscale images of size $1 \times 32 \times 32$
- **Conv1:** $1 \rightarrow 8$ channels, 3×3 kernel, stride=1, padding=1
- **ReLU Activation**
- **MaxPool1:** 2×2 pool, stride=2 ($32 \times 32 \rightarrow 16 \times 16$)
- **Flatten:** $[B, 8, 16, 16] \rightarrow [B, 2048]$
- **FC1:** $2048 \rightarrow 10$ (output logits)

2.1.2 Design Rationale

- **Single Convolutional Layer:** Sufficient for simple digit recognition task
- **8 Feature Maps:** Adequate capacity to capture basic edges and patterns in digits
- **3×3 Kernel:** Standard kernel size balancing receptive field and computational cost
- **Padding=1:** Maintains spatial dimensions after convolution
- **MaxPooling:** Provides spatial invariance and reduces computation

2.2 Model 2: 100-Class Object Recognition

2.2.1 Architecture

- **Input Layer:** Grayscale images of size $1 \times 32 \times 32$
- **Block 1:**
 - **Conv1:** $1 \rightarrow 8$ channels, 3×3 kernel, stride=1, padding=1
 - **ReLU and MaxPool:** 2×2 pool ($32 \times 32 \rightarrow 16 \times 16$)
- **Block 2:**
 - **Conv2:** $8 \rightarrow 16$ channels, 3×3 kernel, stride=1, padding=1
 - **ReLU and MaxPool:** 2×2 pool ($16 \times 16 \rightarrow 8 \times 8$)

- **Block 3:**
 - **Conv3:** $16 \rightarrow 32$ channels, 3×3 kernel, stride=1, padding=1
 - **ReLU and MaxPool:** 2×2 pool ($8 \times 8 \rightarrow 4 \times 4$)
- **Block 4:**
 - **Conv4:** $32 \rightarrow 64$ channels, 3×3 kernel, stride=1, padding=1
 - **ReLU and MaxPool:** 2×2 pool ($4 \times 4 \rightarrow 2 \times 2$)
- **Flatten:** $[B, 64, 2, 2] \rightarrow [B, 256]$
- **FC1:** $256 \rightarrow 100$ (output logits)

2.2.2 Design Rationale

- **Four-Block Feature Funnel:** Uses a deeper architecture to extract highly abstract features necessary for 100-class discrimination.
- **Tapered Channel Expansion:** Starts with a low channel count (8) and doubles it at every spatial reduction ($32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2$) to maintain a balanced computational budget while increasing representational depth.
- **Aggressive Downsampling:** Four pooling layers reduce the final feature map to 2×2 , effectively aggregating global spatial information before the fully connected layer.

3 Trainable Parameters

3.1 Model 1: Parameter Count

Conv1 Parameters:

$$\begin{aligned}\text{Weights} &= C_{out} \times C_{in} \times K_h \times K_w = 8 \times 1 \times 3 \times 3 = 72 \\ \text{Biases} &= C_{out} = 8 \\ \text{Total Conv1} &= 72 + 8 = 80\end{aligned}$$

FC1 Parameters:

$$\begin{aligned}\text{Weights} &= N_{in} \times N_{out} = 2048 \times 10 = 20,480 \\ \text{Biases} &= N_{out} = 10 \\ \text{Total FC1} &= 20,480 + 10 = 20,490\end{aligned}$$

$$\text{Total Parameters (Model 1): } 80 + 20,490 = \boxed{20,570}$$

4 Model 2: Detailed Analysis

4.1 Parameter Count

Conv1 Parameters ($1 \rightarrow 8$):

$$\begin{aligned}\text{Weights} &= 8 \times 1 \times 3 \times 3 = 72 \\ \text{Biases} &= 8 \\ \text{Total Conv1} &= 80\end{aligned}$$

Conv2 Parameters ($8 \rightarrow 16$):

$$\begin{aligned}\text{Weights} &= 16 \times 8 \times 3 \times 3 = 1,152 \\ \text{Biases} &= 16 \\ \text{Total Conv2} &= 1,168\end{aligned}$$

Conv3 Parameters ($16 \rightarrow 32$):

$$\begin{aligned}\text{Weights} &= 32 \times 16 \times 3 \times 3 = 4,608 \\ \text{Biases} &= 32 \\ \text{Total Conv3} &= 4,640\end{aligned}$$

Conv4 Parameters ($32 \rightarrow 64$):

$$\begin{aligned}\text{Weights} &= 64 \times 32 \times 3 \times 3 = 18,432 \\ \text{Biases} &= 64 \\ \text{Total Conv4} &= 18,496\end{aligned}$$

FC1 Parameters ($256 \rightarrow 100$):

$$\begin{aligned}\text{Weights} &= 256 \times 100 = 25,600 \\ \text{Biases} &= 100 \\ \text{Total FC1} &= 25,700\end{aligned}$$

$$\text{Total Parameters (Model 2): } 80 + 1,168 + 4,640 + 18,496 + 25,700 = \boxed{50,084}$$

5 MACs and FLOPs Analysis

5.1 Computational Complexity Definitions

- **MAC (Multiply-Accumulate):** One multiplication followed by one addition
- **FLOP (Floating Point Operation):** Each multiply and add counted separately
- **Relationship:** $\text{FLOPs} = 2 \times \text{MACs}$ (assuming each MAC = 2 FLOPs)

5.2 Conv2D MACs Formula

For a convolution layer:

$$\text{MACs} = H_{out} \times W_{out} \times C_{out} \times (K_h \times K_w \times C_{in}) \quad (1)$$

5.3 Linear Layer MACs Formula

For a fully connected layer:

$$\text{MACs} = N_{in} \times N_{out} \quad (2)$$

5.4 Model 1: MACs/FLOPs per Forward Pass

Conv1:

$$\begin{aligned}\text{MACs} &= 32 \times 32 \times 8 \times (3 \times 3 \times 1) \\ &= 1024 \times 8 \times 9 = 73,728\end{aligned}$$

FC1:

$$\text{MACs} = 2048 \times 10 = 20,480$$

Total Model 1:

$$\begin{aligned}\text{Total MACs} &= 73,728 + 20,480 = \boxed{94,208} \\ \text{Total FLOPs} &= 2 \times 94,208 = \boxed{188,416}\end{aligned}$$

5.5 Model 2: MACs/FLOPs per Forward Pass

Conv1 ($32 \times 32 \times 8$):

$$\begin{aligned}\text{MACs} &= 32 \times 32 \times 8 \times (3 \times 3 \times 1) \\ &= 1024 \times 8 \times 9 = 73,728\end{aligned}$$

Conv2 ($16 \times 16 \times 16$):

$$\begin{aligned}\text{MACs} &= 16 \times 16 \times 16 \times (3 \times 3 \times 8) \\ &= 256 \times 16 \times 72 = 294,912\end{aligned}$$

Conv3 ($8 \times 8 \times 32$):

$$\begin{aligned}\text{MACs} &= 8 \times 8 \times 32 \times (3 \times 3 \times 16) \\ &= 64 \times 32 \times 144 = 294,912\end{aligned}$$

Conv4 ($4 \times 4 \times 64$):

$$\begin{aligned}\text{MACs} &= 4 \times 4 \times 64 \times (3 \times 3 \times 32) \\ &= 16 \times 64 \times 288 = 294,912\end{aligned}$$

FC1 ($256 \rightarrow 100$):

$$\text{MACs} = 256 \times 100 = 25,600$$

Total Model 2 Complexity:

$$\text{Total MACs} = 73,728 + 294,912 + 294,912 + 294,912 + 25,600 = \boxed{984,064}$$

$$\text{Total FLOPs} = 2 \times 984,064 = \boxed{1,968,128}$$

6 Dataset Loading Time

6.1 Dataset Statistics

- **Dataset 1:** 10 classes (digits 0-9), grayscale images resized to 32×32
- **Dataset 2:** 100 classes (CIFAR-100 categories), grayscale images resized to 32×32
- **Preprocessing:** Normalization to $[0, 1]$, data augmentation (random flips, rotation)

6.2 Loading Time Measurements

Metric	Dataset 1	Dataset 2
Training samples	3.7358	4.5397s
Test samples	3.2589	3.7273
Batch size	32	32

Table 1: Dataset loading time measurements

7 Training and Validation Performance

7.1 Training Configuration

- **Optimizer:** Stochastic Gradient Descent (SGD)
- **Learning Rate:** 0.01
- **Batch Size:** 32 (training), 64 (evaluation)
- **Number of Epochs:** 10
- **Loss Function:** Cross-Entropy Loss
- **Data Augmentation:** Random horizontal flip, random rotation ($\pm 10^\circ$)

7.2 Model 1: Training Performance

Epoch	Train Loss	Time (s)
1	1.0697	20.38
2	0.5873	20.12
3	0.5271	21.15
4	0.4729	20.92
5	0.4218	22.04
6	0.3750	21.46
7	0.3359	38.95
8	0.3033	25.66
9	0.2765	21.41
10	0.2547	21.16

Table 2: Model 1 training metrics across epochs

Final Test Accuracy: 92.97% (55785/60000)

7.3 Model 2: Training Performance

Epoch	Train Loss	Time (s)
1	4.5374	129.08
2	4.1907	120.63
3	3.9313	130.99
4	3.7337	151.13
5	3.5913	138.09
6	3.4682	115.92
7	3.3625	118.03
8	3.2729	118.07
9	3.1994	119.53
10	3.1331	120.73

Table 3: Model 2 training metrics across epochs

Final Test Accuracy: 27.53% (13765/50000)

8 Performance Visualization

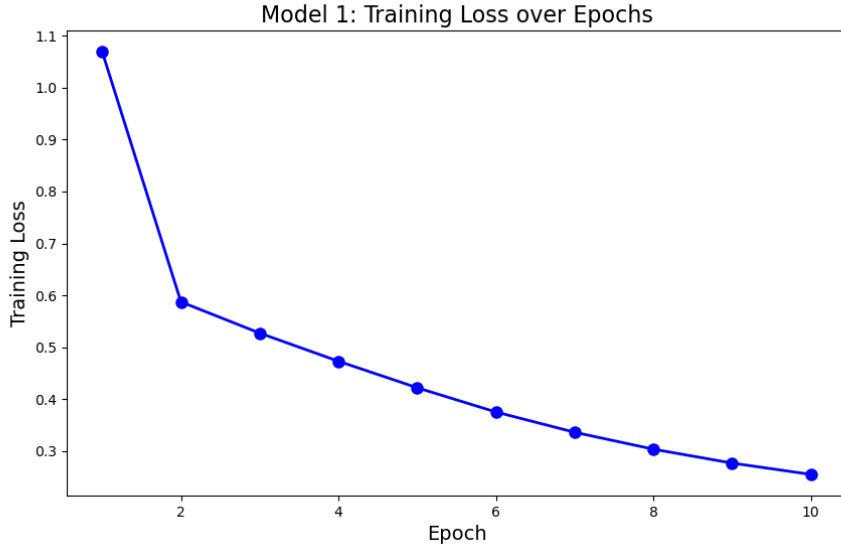


Figure 1: Model 1: Training loss over epochs

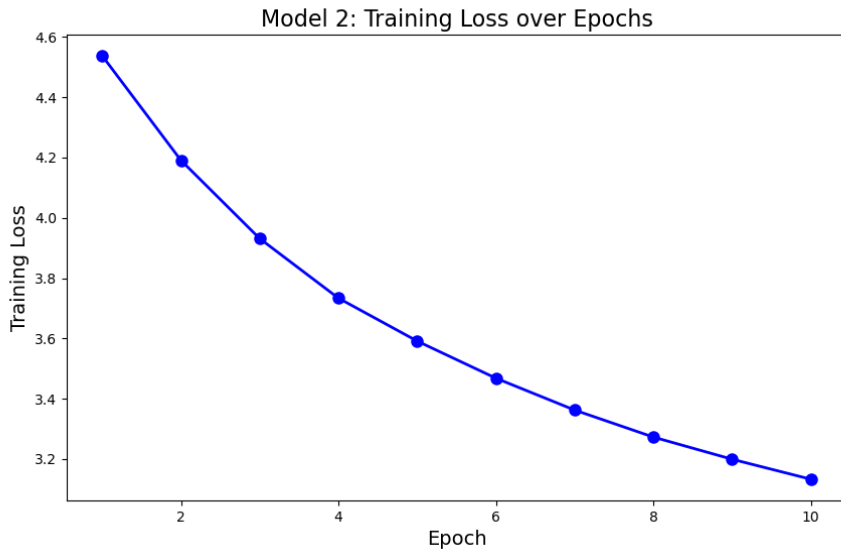


Figure 2: Model 2: Training loss over epochs

9 Failed Design Decision

9.1 Attempted Design: Three Convolutional Layers

9.1.1 Initial Architecture

I initially attempted a deeper architecture for Model 2 with three convolutional layers:

- Conv1: $1 \rightarrow 16$ channels, MaxPool ($32 \rightarrow 16$)
- Conv2: $16 \rightarrow 32$ channels, MaxPool ($16 \rightarrow 8$)
- Conv3: $32 \rightarrow 64$ channels, MaxPool ($8 \rightarrow 4$)
- FC: $64 \times 4 \times 4 = 1024 \rightarrow 100$

9.1.2 Why It Failed

1. Vanishing Spatial Information

- After three maxpool operations, spatial dimensions reduced to 4×4
- Original 32×32 images already small; excessive downsampling lost critical details
- Fine-grained features needed for discriminating 100 classes were destroyed

2. Overfitting

- Training loss decreased rapidly but test accuracy remained poor
- High capacity but insufficient training data
- Model memorized training samples rather than learning generalizable features